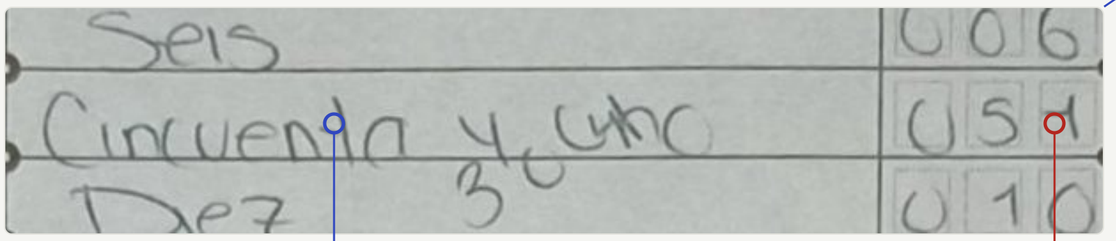
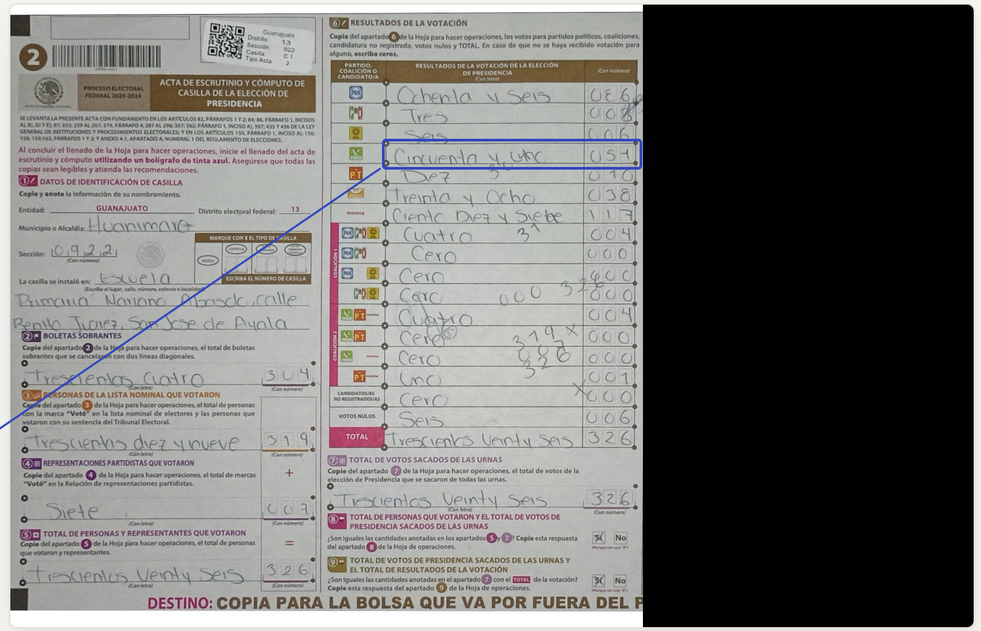
AMD Developer Hackathon ACT III

Two Hands

Verifiable quality control for handwritten numbers.

OCR reads the number. Two Hands checks whether the whole form agrees.





Tally sheet 110922C0100 (INE 2024), privacy-masked: the names-and-signatures column is blacked out.

The paper: 51
the words: cincuenta y uno; certified count: 51

Gemma read 54
digits and words; rows sum to 329, TOTAL 326: routed

1,270
real handwritten tally sheets read by Gemma on AMD

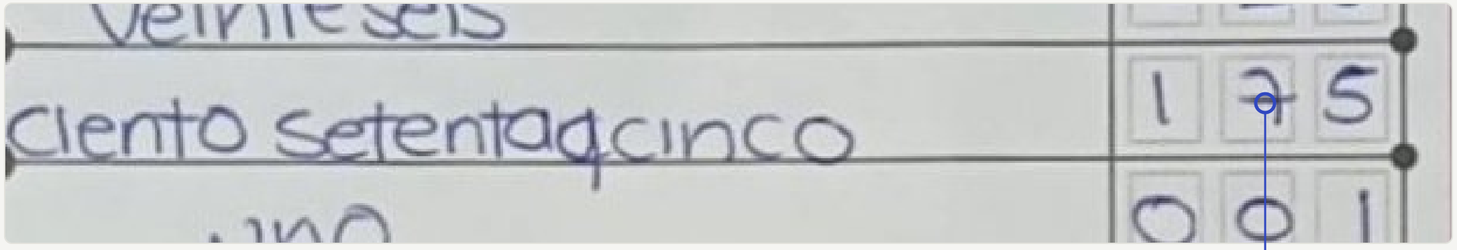
91.6%
field match, vs 62.84% for the reading the state bought

27 of 28
certified-detectable capture errors caught

$1
per sheet audited (proposed)

THE PROBLEM, FROM INE'S OWN FILES

Two official records of the same paper disagree. Neither says which one is right.



The paper: 175
in digits and in words

Form 050815B0100, Option 07 (masked, cropped right of the logo column). One of the 52 on our random 300.

THE TWO TYPED RECORDS OF THIS FIELD

Typed into the preliminary count	~~75~~
Certified count	175

The 75 is a typing error in the database, not on the paper.

6.5%

of the 52,167 polling stations not recounted: the preliminary capture and the certified count disagree on at least one vote field

19.2%

of nationally captured tally forms fail their own row sum (26,336 of 137,196)

Double keying copies the paper faithfully but **cannot say which paper is wrong**. The photo can.

HOW IT WORKS

The model reads. The paper's own arithmetic decides.

01
Photo
Page located and privacy-masked before it is stored or sent

02
Gemma reads
Gemma-4-31B-IT on AMD Token Factory reads every field twice: digits and words

03
Arithmetic
words = digits · rows sum to TOTAL · section 5 = section 7 = TOTAL

04
Accept or route
All hold: **accepted**. One fails: **a person** gets the field, with its photo

05
Receipt
Each field a Merkle leaf; the root anchored with OpenTimestamps

211298B0100

INE 2024 tally sheet, type 2 · phone photo · Puebla · certified by comparison

sample: capture differs from certified

Accepted: 0 fields to key
Every identity on the page holds, so nobody keys this form.

✓	rows sum to TOTAL	417 = 417
✓	votaron + representantes = section 5	417 = 417
✓	section 5 = section 7	417 = 417
✓	section 7 = TOTAL	417 = 417
✓	words = digits on every field	23 of 23

An accepted form on the site (211298B0100): every identity holds, so nobody keys it. A second reader, Qwen3.8-27B on AMD, never overrides Gemma: it can only withhold acceptance.

ONE REAL SHEET, REQUEST TO RECEIPT

Gemma read 23. The paper says 123. The sum caught it.

IDENTITY	LEFT SIDE	RIGHT SIDE	RESULT
rows sum to TOTAL	**366** Option 01 1 + Option 02 23 + Option 03 3 + Option 04 26 + Option 05 22 + Option 06 98 + Option 07 165 + Option 08 0 + Option 09 0 + Option 10 0 + Option 11 0 + Option 12 7 + Option 13 0 + Option 14 0 + Option 15 0 + Unregistered candidates 0 + Null votes 21	**466** TOTAL 466	FAIL
votaron + representantes = section 5	**466** Voters on the list 466 + Representatives who voted 0	**466** Voters + representatives 466	pass
section 5 = section 7	**466** Voters + representatives 466	**466** Ballots taken from the boxes 466	pass
section 7 = TOTAL	**466** Ballots taken from the boxes 466	**466** TOTAL 466	pass

/trace on the site: the gate re-run on Gemma's stored raw output for form 122655C0100.

THE READ

23 in digits and 23 in words: they agree, so only the arithmetic can catch it.

THE GATE

Rows sum to 366, the form's TOTAL is **466**: routed to a person. With the paper's 123: 466 = 466.

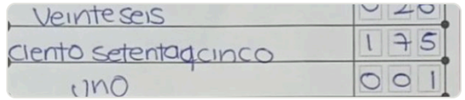
THE RECEIPT

VERIFIED: the record is unchanged since the report was sealed. Not that the reading is right.

RESULTS · PRE-REGISTERED · INE'S PUBLIC DATA

1,270 real tally sheets read on AMD. 49 capture errors confirmed by the certified count.

3



In English: words: "ciento setenta y cinco" = 175 · boxes: 1 7 5

050815B0100 · Option 07 STRONG

receipt

Judge this verdict yourself

Keyed	Model digits	Model words
75	175	175

- model digits and words agree (175)
- the keyed value breaks rows sum to TOTAL (258 vs 358); the model's value satisfies it

One dispute in the audit report: the crop, the keyed 75, Gemma's 175 in digits and words.

RANDOM 300, PRE-REGISTERED

52 fields where the official capture does not match the paper, in 6,358 checked; each with its photo.

COLIMA, A WHOLE STATE

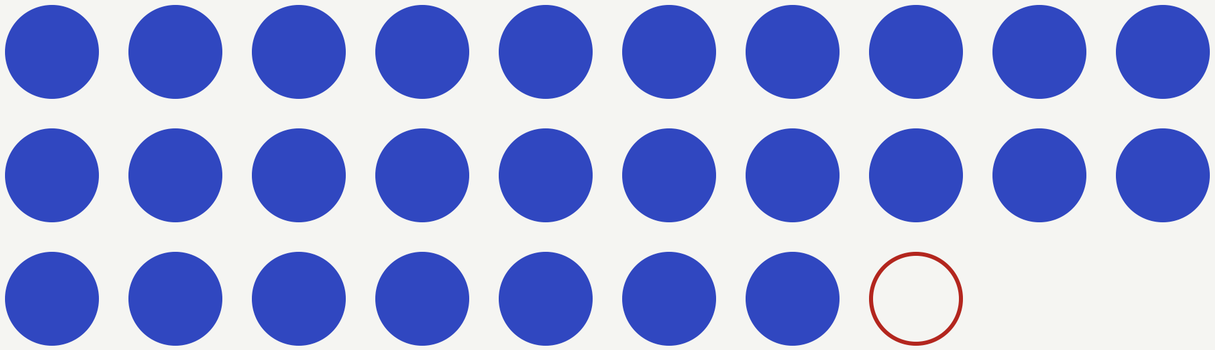
All **982** located presidential sheets for **$0.14**; 27 fields where the capture does not match the paper, each confirmed by the certified count.

100 FRESH FORMS

Gate v2 accepted 22 forms: **0 of 484** fields wrong.

CHECKED

27 of 28 capture errors caught. A blind re-read found 0 differences.



Each dot: a capture error the certified count reveals in Colima. Filled: caught. Outlined: missed.

RECALL

27 of 28, 96% (Wilson 95% 82-99%). One state, pre-registered; certified-detectable errors only, not total recall.

A PERSON'S BLIND RE-READ

of the AI-judged disputes: **0 read differently**.

AGAINST THE INCUMBENT · BAJA CALIFORNIA SUR, 2024 LOCAL ACTAS

91.6% field match, where the reading the state bought scored 62.84%.

Two Hands
Gemma on AMD · random 300 actas, pre-registered



91.6%

Google Document AI
the state's purchased reading · all 2,020 actas



62.84%

Same per-field exact-match metric. Ours: 3,622 fields, Wilson 95% 90.63-92.44%, against the published PREP values. Theirs: 14,501 of 23,077 fields, against the manual capture.

GEMMA ON AMD TOKEN FACTORY

Same 300 forms, same gate: Gemma accepts 1.8× the forms Qwen does, 4.6× cheaper, 2.8× faster.

READER, ON AMD	FORMS ACCEPTED	WRONG OF ACCEPTED FIELDS	$ PER 1,000 FORMS	TIME PER READ
Gemma-4-31B-IT	**55 of 300**	**2 of 1,210**	**$0.14**	**87 s**
Qwen3.8-27B	31 of 300	1 of 682	$0.64	241 s
Both must agree, the paper's sums as referee	89 of 300	**0 of 1,958**	$0.78	–

Two readers: lighter gate, a secondary analysis. Qwen: +2 undecided. Source: app/data/bakeoff_300.json.

4.8 forms / min

$0.000146 per form, shared endpoint, 4 in flight

$0.14

Colima, a whole state: all 982 located presidential sheets

150 of 150

Evolus "Audit finding" runs completed; the agent runs Gemma on AMD

BUSINESS AT A GLANCE

An independent check of the keyed values. $1 per sheet audited (proposed).

Audit report: 6,358 fields checked, 267 disputed, each with its photo

PREP 2024, 300 pre-registered forms · 300 forms in the keyed file, 289 compared · audit `ab2b0555076b1` · 2026-10-17 · Merkle root `e56a8ef4f011b9e5` · fingerprint `c4788f2501814fd1`

Summary

Fields checked (keyed value vs the model's read of the paper)	**6,358**
Disputed fields (the paper, as read, disagrees with the keyed value)	**267**
with strong evidence that the keyed value is wrong	**18**
with medium evidence	**80**
with weak evidence (the model may have misread)	**169**

The product: the audit report, every dispute with its photo, sealed by one Merkle root.

BUYER

INE and the 32 state institutes, each with an outside auditor.

WHY NOW

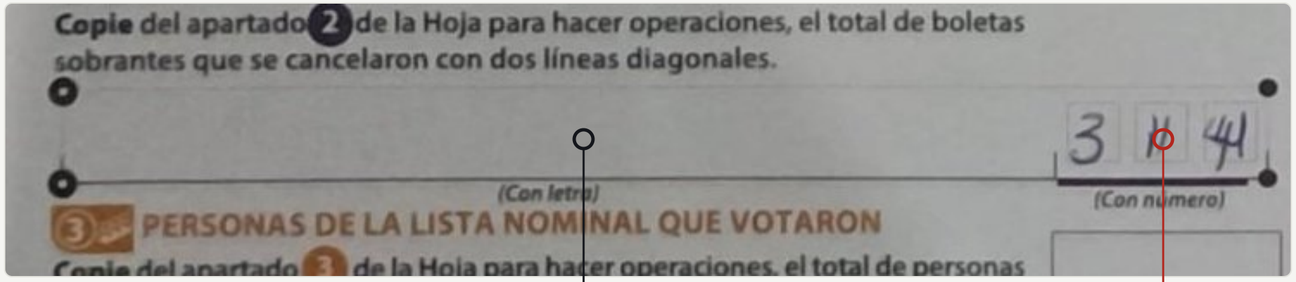
6 June 2027: the federal midterm and 32 concurrent local elections. First step: a **paid shadow-mode pilot** on one state's count.

ONE STATE, PRICED

Morelos paid MXN 26.6M to run its 2024 PREP. Our check of every sheet: about **MXN 143k**, 0.5% of that.

WHERE IT HOLDS, AND WHERE IT BREAKS

Read 3114, paper 314. Gate v2 sends it to a person.



Words line: blank
no second reading to compare

The model read 3114
the paper says 314

Form 070212B0100, unused ballots (masked crop).

THE BLIND SPOT

Unused ballots sit in no identity: no sum can catch a digit inserted here.

HELD OUT

On 100 fresh forms the frozen gate let it through. Gate v2, digits and words both read and equal, blocked it: **0 wrong of 484**.

SCOPE

Clerical capture quality only. Most forms still go to a person. Not an official INE product.

CHECK IT YOURSELF

Every number re-derives from public data.

REPRODUCE

```
make eval
```

Recomputes every number offline and checks it against what the site shows.

VERIFY A RECEIPT

```
python -m twohands.verify_receipt
  receipt.json --crop crop.jpg
```

A changed value or crop fails.

CHECK THE TIMESTAMP

```
ots verify
results/sample_audit_root.txt.ots
```

The report's Merkle root, anchored in Bitcoin with OpenTimestamps.

Two Hands. Verifiable quality control for handwritten numbers. `github.com/py-tr/twohands`